NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE

(AN AUTONOMOUS INSTITUTION)

MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCDSMT301

Course Name : Data Science & Machine Learning

Max. Marks: 60 Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	Compare descriptive statistics for univariate vs. multivariate datasets.	4
2	Differentiate between filter, wrapper, and embedded methods.	4
3	Explain the concept of Ordinary Least Squares (OLS) estimation in regression. How is it used to minimize errors? Illustrate with a small numerical example showing calculation of residuals and squared error.	4
4	Define the term margin in SVM. How does increasing the margin affect the generalization ability of the classifier? Support your answer with a diagram.	4
5	Define Precision, Recall, and F-measure. A classifier has Precision = 0.90 and Recall = 0.75. Calculate the F1-score and explain in which situations F1-score is more informative than accuracy.	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	What are the common types of data visualizations? Give examples.	8
	OR	
B	List and explain the key stages of the Data Science Process Cycle. For each stage, give a practical example.	8
C	Classify a new person P with attributes: AGE = 24, GENDER = 0 (Male) using k = 3 and Euclidean distance. Show all distance calculations and determine the class.	

NAME	AGE	GENDER (0=Male1=Female)	CLASS OF SPORTS
Aryan	18	0	Football
Priya	25	1	Cricket
Karan	30	0	Neither
Anika	20	1	Football
Sameer	35	0	Cricket
Riya	22	1	Cricket
Aditya	40	0	Neither
Neha	19	1	Football
Dev	28	0	Cricket
Meera	15	1	Neither

	OR	
D	Predict whether PlayTennis = Yes or No for this instance (Outlook = Overcast, Temperature = Cool, Humidity = High, Windy = True). Show all calculations including: • Prior probabilities • Conditional probabilities with Laplace smoothing • Joint probability computation **Outlook** | **Temperature** | **Humidity** | **Windy** | **PlayTennis** Sunny | Hot | High | False | No Sunny | Hot | High | True | No Overcast | Hot | High | False | Yes Rainy | Mild | High | False | Yes Rainy | Cool | Normal | False | Yes Rainy | Cool | Normal | True | No Overcast | Cool | Normal | True | Yes Sunny | Mild | High | False | No Sunny | Cool | Normal | False | Yes Rainy | Mild | Normal | False | Yes Sunny | Mild | Normal | True | Yes Overcast | Mild | High | True | Yes Overcast | Hot | Normal | False | Yes Rainy | Mild | High | True | No	8
E	a) Explain the difference between pre-pruning and post-pruning, including one advantage and disadvantage of each. b) A decision tree built on a training set achieves 95% accuracy but only 70% accuracy on the test set. Discuss how pruning could help in this situation and why overfitting occurs.	8
	OR	
F	A company wants to predict employees' salary based on working hours (X1) and experience (X2). The regression equation obtained is: $Y=20000+1500X1+2500X2$ a) Interpret each coefficient and the intercept. b) Predict the salary for an employee working 160 hours and having 5 years of experience. c) Discuss the impact of high correlation between X1 and X2 (multicollinearity) on the reliability of the regression coefficients.	8
G	a) Explain the role of activation functions in neural networks. Compare Sigmoid, Tanh, and ReLU activations in terms of their range, gradient behavior, and impact on network training. b) For a 2D dataset, explain how a hyperplane separates two classes. Illustrate with a diagram how the maximum margin hyperplane is different from other possible separating hyperplanes.	8
	OR	

H	a) Describe the different network topologies in neural networks (single-layer feedforward, multi-layer feedforward, recurrent). Give one application example for each. b) Explain the concept of a support vector. Why are only the support vectors important in defining the decision boundary, while the other data points are not?	8
I	A medical test is used to detect a rare disease. Out of 1,000 patients, only 50 actually have the disease. The classifier predicts 60 patients as positive, out of which 45 are correct. a) Construct the confusion matrix. b) Calculate Accuracy, Precision, Recall (Sensitivity), and Specificity. c) Explain why accuracy is misleading in this case and why Precision, Recall, or F1-score are better measures for evaluating the model.	8
	OR	
J	Compare Bagging, Boosting, and Random Forests in terms of working principle, handling overfitting, and suitability for different datasets. Provide one real-life application for each.	8

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MCA Degree S3 (Regular) Examination October 2025

Course Code: 24MCDSMT301

Course Name : DATA SCIENCE & MACHINE LEARNING

Max. Marks: 60

Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	What is meant by structuring data? Why is it important?	4
2	Why is feature selection necessary before building a machine learning model?	4
3	Explain rule based classification and how is this technique used to derive classification rules from decision tree	4
4	Explain the strengths and weaknesses of training neural networks with back propagation?	4
5	Explain why K-Means may fail for non-spherical clusters. Suggest two methods to improve the performance of K-Means clustering	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	With the help of diagram explain data science process.	8
	OR	
B	Discuss the following multivariate visual exploration techniques with a neat plot (a) Scatter Plot (b) Bubble Chart (c) Density Chart (d) Quartile Plot	8
C	Classify a new person P with attributes: AGE = 23, GENDER = 0, HEIGHT = 172 cm, WEIGHT = 66 kg, using k = 3 and Euclidean distance. Show all calculations.	8

NAME	AGE	GENDER (0=Male,1=Female)	HEIGHT(cm)	WEIGHT(kg)	Class of Sports
Ajay	22	0	170	65	Football
Sara	19	1	160	55	Cricket
Rahul	25	0	175	70	Cricket
Pooja	21	1	165	60	Football
Mark	30	0	180	75	Neither
Anika	20	1	162	58	Cricket
Rohan	28	0	178	72	Cricket
Tina	18	1	158	54	Football

OR

D Using Naive Bayes with Laplace smoothing, predict PlayTennis for a new instance:

Outlook = Sunny, Temperature = Cool, Humidity = High, Windy = True.

Show all calculations, including:

- Prior probabilities with Laplace smoothing
- Conditional probabilities with Laplace smoothing
- Joint probability and final classification

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

E Consider the training dataset.

Outlook	Temp	Humidity	Wind	Play Tennis(Target Feature)
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes

a) Find the entropy of the training dataset with respect to target feature 'Play Tennis'.

b) Choose the best attribute to split the training dataset.

	OR	
F	Explain the Divide and Conquer approach in Decision Tree learning with a suitable example. How does it differ from the Separate and Conquer approach used in classification rule learning?	8
G	Explain the back propagation algorithm used for training neural networks. Discuss the role of learning rate and error minimization in this process.	8
	OR	
H	With a neat diagram, explain how support vectors are identified and how they help in determining the maximum margin hyperplane in SVM.	8
I	Explain the K-means clustering algorithm with an example. Discuss how the choice of distance measure and the number of clusters (k) affects the quality of clustering.	8
	OR	
J	A binary classifier predicts whether patients have a disease. Construct a confusion matrix for the following case: • 90 Patients actually sick, 10 actually healthy. • Classifier predicted: 85 sick correctly, 5 sick incorrectly as healthy, and 2 healthy incorrectly as sick. From this, calculate Accuracy, Precision, Recall (Sensitivity), and Specificity.	8

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NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE

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MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCDAAT302

Course Name : DESIGN AND ANALYSIS OF ALGORITHMS

Max. Marks: 60 Duration: 3 Hrs

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PART A

*Answer **all questions**. Each question carries 4 marks (One questions from each module)*

		Marks
1	Quick sort algorithm has worst case time complexity of O (n^2). Justify your answer	4
2	Define the term minimum spanning tree with an example.	4
3	Define the term lower bound theory? Give an example for lower bound.	4
4	Explain the properties of flow network	4
5	Explain the significance of randomized algorithm. Mention its use.	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	What is meant by asymptotic complexity of an algorithm? Explain the three notations used for representing asymptotic complexity?	8
	OR	
B	Explain how divide and conquer strategy is used in Merge sort.	8
C	Explain the greedy strategy with the control abstraction model.	8
	OR	
D	Explain the travelling salesman problem using Dynamic programming.	8
E	Demonstrate solving the 8-puzzle problem using branch and bound.	8
	OR	
F	Explain how state space trees are built in backtracking, using the 4-queens problem as an example.	8
G	Differentiate between class P and class NP with respect to verifiability vs solvability.	8
	OR	
H	Prove the max-flow min-cut theorem with an example network diagram.	8
I	Compare and contrast merits and demerits of randomized algorithms over deterministic ones in large-scale computation.	8

	OR	
J	Design a step-by-step vertex cover approximation algorithm for an undirected graph. Mention its significance in algorithm strategy.	8

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NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE

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MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCCCOT303

Course Name :CLOUD COMPUTING

Max. Marks: 60 | Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	Differentiate between private cloud and public cloud	4
2	Explain asymmetric clustering and symmetric clustering.	4
3	Differentiate between Ephemeral storage and persistent storage	4
4	Discuss on Tunnel-based networks and its components	4
5	Explain MMM replication	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	Compare the different components in OpenStack logical Architecture	8
	OR	
B	Discuss on keystone identity management in Open Stack	8
C	Describe the functionalities between Federated keystone and fernet keystone services	8
	OR	
D	Differentiate between nova conductor service and nova scheduler services	8
E	Describe The Swift ring with diagram	8
	OR	
F	Explain Share storage service of Manila with diagram	8
G	Explain Linux-bridge based Connectivity	8
	OR	
H	Distinguish between VLAN-based and tunnel- based networks	8
I	List the HA in database and its levels with diagram	8
	OR	
J	Differentiate between HA in database and queue	8

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MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCCCOT303

Course Name : CLOUD COMPUTING

Max. Marks: 60 | Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	Differentiate between public, private, and hybrid cloud architectures. Give one real-world example for each	4
2	Compare symmetric and asymmetric clustering in an OpenStack environment.	4
3	Describe the primary components of the OpenStack Compute service and their roles in managing virtual instances.	4
4	Detail the function of the ML2 Plugin and its significance in implementing virtual networks in OpenStack.	4
5	Explain the purpose of the Heat service and its components in OpenStack orchestration.	4

PART B

	*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*	
A	Discuss the logical architecture of OpenStack and the critical architectural considerations for designing an OpenStack cloud.	8
	OR	
B	Describe and compare the three main cloud service models. How these services collectively form the foundation of cloud computing. Explain.	8
C	Explain the purpose of the OpenStack API services. How do these services enable interaction with various OpenStack components?	8
	OR	
D	Discuss the importance and key functionalities of the OpenStack Horizon dashboard, explaining how it simplifies cloud resource management for both end-users and administrators.	8
E	Compare and contrast the three types of OpenStack Storage: Block, Object, and File Share. For each type, describe a suitable use case and explain the key	8

	characteristics that make it a good fit for that purpose.	
	OR	
F	Describe the architectural design and purpose of the OpenStack Swift service, explaining how it provides object storage capabilities and highlighting its key advantages for storing and retrieving unstructured data.	8
G	Detail the role of floating IPs in OpenStack Networking, describing how they are associated with virtual machines and how this mechanism enables connectivity from the external world to instances within a private network.	8
	OR	
H	Explain the architecture of the OpenStack Neutron service. Enumerate how it provides networking capabilities to virtual instances and how its various components, such as virtual switches, work together to achieve this.	8
I	Describe the scope of High Availability (HA) in an OpenStack environment, explain its importance for cloud services, and detail how HA is implemented in the OpenStack database and Queue.	8
	OR	
J	Outline the concept of orchestration in OpenStack and its benefits for managing cloud resources, and compare OpenStack Heat with Terraform by highlighting their key features and usage.	8

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MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCCLFT304

Course Name : CYBER LAWS AND FORENSICS

Max. Marks: 60 | Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	Discuss about protection of copyright on cyberspace	4
2	Explain about intermediaries	4
3	What is AFF?	4
4	Discuss about the importance of Windows Registry	4
5	What are the main considerations in disqualification	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	Describe ISP in cyberspace and cyberspace and protection of patents in India	8
	OR	
B	Describe the importance of Cyber laws	8
C	List out Amendments on IT Act	8
	OR	
D	Explain about Information Technology Act and Punishments	8
E	Discuss detail about Acquisition tools and methods	8
	OR	
F	Discuss detail about Evidence Retention and Familiarizing Autopsy for Windows	8
G	Explain the structure and features of Microsoft Bit Locker.	8
	OR	
H	List out Microsoft file systems.	8
I	Describe Guidelines for writing reports and also explain about Layout and presentation of reports	8
	OR	
J	Describe details about importance of Forensic Reports and its features.	8

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QP CODE : MC0325CLFPMD

Total Pages: 01

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MCA Degree S3 (Regular) Examination October 2025

Course Code : 24MCCLFT304

Course Name : CYBER LAWS AND FORENSICS

Max. Marks: 60 Duration: 3 Hrs

PART A

*Answer **all questions**. Each question carries 4 marks*
(One questions from each module)

		Marks
1	Write the Importance of cyber laws	4
2	Explain the punishment for violation of privacy.	4
3	How do you Conduct an investigation	4
4	Write the Importance of Disk Drives and its usage	4
5	Explain about Autopsy.	4

PART B

*Answer **any 5 full questions** with minimum one question from each module. Each question carries 8 marks*

A	List out the main points deception by squatting in cyberspace	8
	OR	
B	List out differences between Linking, hyperlinking and framing.	8
C	Describe punishment for Abetment and Attempt to commit offences under the IT act.	8
	OR	
D	Explain about Information Technology Act and Punishments	8
E	Discuss detail about Digital evidence validation methods and tools	8
	OR	
F	What are the Storage formats for storing collected digital evidence	8
G	Discuss detail about Microsoft File Systems and its features	8
	OR	
H	Discuss details about Microsoft BitLocker	8
I	Describe Guidelines for writing reports and also explain about Layout and presentation of reports	8
	OR	
J	Explain about Ethical difficulties and ethical responsibilities in creating Forensic report	8

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